GG-A-438a
JULY 30, 1956
SUPERSEDING
Int. Fed. Spec. GG-A-00438a (GSA-FSS)
February 23, 1956

FEDERAL SPECIFICATION

ALBUMINOMETER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers albuminometers that are used to ascertain the quantity of protein (albumens) in a liquid.

1.2 Classification.

1.2.1 *Type, grade and class.*—Albuminometers shall be of but one type, one grade and one class.

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

ZZ-S-751—Stoppers, Rubber.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Glass.*—Tube shall be made of good quality corrosion-resisting glass, of such composition and so annealed that the albuminometers will meet the requirements specified hereinafter and withstand the tests described in 4.3.

3.1.2 *Rubber.*—Stopper shall be made of rubber, or suitable synthetic rubber, conforming to the applicable requirements of type A, grade I, of Federal Specification ZZ-S-751.

3.1.3 *Wood.* — Wood shall be hardwood species, of good quality, suitable in all respects for the purpose intended.

3.2 Instructions. — Clear and adequate printed instructions for use shall be furnished with each albuminometer.

3.3 Description.—The albuminometer outfit shall be the Esbach type and consist essentially of a graduated glass tube, a stopper of rubber or suitable synthetic substitute for rubber, a wooden base, and a wooden cover.

3.4 Physical requirements.

3.4.1 *Graduated glass tube.* — One glass tube shall be furnished with each albuminometer. The tube shall comprise as integral parts, a graduated tubular section tapering on the lower portion and a glass peg fused onto the tapered end. The complete tube shall

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be approximately 155 mm. over-all length, and 21 mm. outside diameter except for the lower portion, which shall be tapered for a distance of approximately 50 mm. The peg shall be approximately 25 mm. long and approximately 7 mm. outside diameter, fused to the end of the tapered portion of the tube. The open end of the tube shall be cut square with the length, shall be provided with a small reinforcing bead, and shall be fire-polished and smoothly finished.

3.4.1.1 *Solubility.*—When tested as outlined in paragraph 4.3.1, no water mark shall remain upon the inner walls of the tubes, nor pink color develop in the testing solution.

3.4.1.2 *Boiling.*—The tubes shall show no evidence of corrosion, scumming, chipping, or cracking, when tested as outlined in paragraph 4.3.2.

3.4.1.3 *Thermal shock.*—The tubes shall withstand, without cracking, the test outlined in paragraph 4.3.3.

3.4.1.4 *Graduations and markings.* — The tube shall be accurately graduated from the bottom up to 2-ml. capacity in $\frac{1}{10}$-ml. graduations, from 2 to 10-ml. capacity in $\frac{1}{2}$-ml. graduations, with a further graduation at the 20-ml. capacity. A plus or minus tolerance of $\frac{1}{10}$ ml. for the interval 0 to 2 ml. and a plus or minus tolerance of $\frac{1}{4}$ ml. for the interval 2 to 20 ml. will be allowed. The graduations shall be perpendicular to the long axis and parallel with each other. The $\frac{1}{10}$-ml. and $\frac{1}{2}$-ml. capacity graduation markings shall be not less than 3 mm. long. The $\frac{1}{2}$-ml. and $1\frac{1}{2}$-ml. capacity graduation marks and the main 1-ml. capacity graduation markings shall be longer than the sub-graduation markings. Each 1-ml. capacity graduation shall be numbered with Arabic numerals on the right of the scale. At the left of the graduation indicating 10 ml. shall be placed the capital letter "U", and at the left of the line indicating 20 ml. shall be placed the capital letter "R". The graduation markings shall be clean cut, legible, and filled with a permanent color, or fused-in color which shall withstand the test for permanency of pigment outlined in Section 4. The lines and numbers of tested samples, when examined visually, shall present the same appearance as before testing.

3.4.2 *Stopper.*—One stopper, size 2, conforming to the applicable requirements of Federal Specification ZZ–S–751, shall be furnished with each albuminometer.

3.4.3 *Base.*—A removable hardwood base, of adequate size for use as a stand for the graduated glass tube, shall be provided with each albuminometer. The base shall have a hole concentrically placed, to receive the glass peg on the tube which shall hold the loaded tube in an upright position. The upper portion of the base shall be so designed that it will allow the cover to fit snugly down over it, holding the cover in an upright position.

3.4.4 *Cover.*—One hardware cover shall be provided with each albuminometer. The cover shall be cylindrical in shape, closed on the upper end. The inside diameter shall be of adequate size to encase the graduated glass tube with stopper, and fit in proper position on the base.

3.5 **Workmanship.**—Workmanship shall be first class throughout. Albuminometers shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 **Sampling.**—Not less than 2 albuminometers shall be taken at random from each 100 or fraction thereof delivered, for inspection and tests.

4.2 **Inspection.**—Samples shall be carefully inspected for defects in material, workmanship, markings, dimensions, finish, and other applicable requirements.

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4.3 Tests.

4.3.1 *Solubility.*—Not less than 50 percent of the albuminometers taken for test (4.1) shall be well cleaned and then rinsed three times with distilled water, using a new quantity of distilled water for each rinsing. The albuminometers shall be air dried in an inverted position for 1 minute after each rinsing. After the final drying operation the albuminometers shall be half filled with distilled water which has been boiled for 10 minutes in a flask (which in itself will not affect the test solution) containing 0.2 percent U. S. P. phenolphthalein test solution and autoclaved under 15 pounds pressure for 30 minutes.

4.3.2 *Boiling test.*—The albuminometers taken for test shall be completely immersed in distilled water in a closed vessel having a small steam vent and boiled continuously for 6 hours, boiling water being added when necessary, to compensate for evaporation losses. The albuminometers shall be suspended in the water to avoid contact with the containing vessel.

4.3.3 *Shock test.* — The albuminometers shall be immersed in tap water at 18.5°C., plus or minus 2½°C. (65°F., plus or minus 5°F.) allowed to remain 10 minutes, then transferred suddenly into boiling water. This operation shall be repeated five times.

4.3.4 *Permanency of pigment.*—Shall be tested as follows:

4.3.4.1 *Alkali test.*—Immerse samples completely in boiling normal sodium hydroxide, cover, and boil 1½ hours. During boiling period make up evaporative losses with boiling distilled water. Do not permit sodium hydroxide to concentrate itself more than 5 percent over the initial strength.

4.3.4.2 *Acid test (hydrochloric).*—Immerse samples completely in boiling concentrated hydrochloric acid (sp. gr. 1.18) and let boil for 1½ hours, replacing loss by fresh concentrated acid. During boiling do not let the volume decrease more than 5 percent before adding fresh acid.

4.3.4.3 *Acid test (sulphuric).* — Immerse samples completely in concentrated sulphuric acid at room temperature. Let stand 1 hour.

4.3.4.4 At end of each test period remove samples and rinse thoroughly in distilled water. Dry by rubbing vigorously with a cloth.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Packages.*—Each cardboard box or carton shall be plainly labeled with the name of the item, quantity contained, and the name of the manufacturer or contractor.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the quantity contained therein, as defined by the contract or order under which the shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

6. NOTES

6.1 Purchasers should exercise any desired options offered herein and specify class required. (See paragraphs 5.1, 5.2, and 5.3.2).

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6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 Transportation description.—Transportation description applicable to this item is:

Glassware, chemical laboratory
Carload minimum weight 18,000 pounds, subject to Rule 34, Uniform Freight Classification

Glassware, laboratory
Truckload minimum weight 18,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
ARMY: M.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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